

Model Predictive Control (MPC)

Optimization and Learning for Robot Control

RCI Lab

Kyung Hee University

February 17, 2026

1. What is Model Predictive Control (MPC)?

Core Idea (Receding Horizon Control): Instead of finding a single global control policy, we solve an Optimal Control Problem (OCP) at **every time step** over a finite horizon.

The MPC Loop:

- 1 Measure current state x_t .
- 2 Solve Finite-Horizon OCP:

$$U^* = \arg \min_U \sum_{k=0}^{N-1} l(x_k, u_k) + l_f(x_N)$$

$$\text{s.t. } x_{k+1} = f(x_k, u_k), \quad x_0 = x_t$$

- 3 Apply **only the first control input** $u_t = u_0^*$.
- 4 Wait for next time step $t + 1$, shift the horizon, and **repeat**.

2. Prediction vs. Reality

The Horizon Shift Effect:

- At time t , we optimize for $[t, t + N]$.
- At time $t + 1$, we optimize for $[t + 1, t + 1 + N]$.

Even without external disturbances, the predicted trajectory $x_{k|t}$ and the actual trajectory usually differ because the optimization horizon changes (We see "further" into the future at $t + 1$).

Key Challenges in MPC:

- 1 **Feasibility:** Will the OCP always have a solution? (Recursive Feasibility)
- 2 **Stability:** Does applying u_0^* drive the system to the goal?
- 3 **Computation:** Can we solve the OCP within milliseconds?

3. Stability: Infinite vs. Finite Horizon

Infinite Horizon ($N = \infty$): If we solve for $N = \infty$, the Value Function $V^*(x)$ naturally acts as a **Lyapunov Function**, guaranteeing stability (Bellman's Principle).

Finite Horizon ($N < \infty$): Truncating the horizon breaks the Lyapunov property. The "greedy" short-term strategy might lead to a dead-end.

Fix: Terminal Ingredients We approximate the "tail" ($N \rightarrow \infty$) using:

- **Terminal Cost** $l_f(x)$: Approximates cost-to-go from N to ∞ .
- **Terminal Constraint** $x_N \in \mathcal{X}_f$: Ensures we end up in a "safe" region.

4. Sufficient Conditions for Stability

To guarantee stability with finite N , we need three conditions on the Terminal Set $\mathcal{X}_f$ and Cost $l_f(x)$:

- 1 **Control Invariance:** Once in $\mathcal{X}_f$, there exists a control $u = \kappa_f(x)$ that keeps the system in $\mathcal{X}_f$.

$$\forall x \in \mathcal{X}_f, \quad f(x, \kappa_f(x)) \in \mathcal{X}_f$$

- 2 **Cost Descent (Lyapunov Condition):** Inside $\mathcal{X}_f$, the terminal cost must decrease sufficiently.

$$l_f(f(x, \kappa_f(x))) - l_f(x) \leq -l(x, \kappa_f(x))$$

- 3 **Local Validity:** $l_f(x)$ matches the infinite horizon cost locally (e.g., solve LQR at the goal).

Result: $V_N^*(x)$ becomes a valid Lyapunov function $\implies$ **Stability**.

5. Practical Stability Strategies

Calculating the perfect $\mathcal{X}_f$ and l_f is hard for nonlinear systems.

Common Approaches in Robotics:

- **Option 1 (Zero Terminal Constraint):** Force $x_N = x_{goal}$.
 - Pros: Theoretically stable.
 - Cons: Small basin of attraction (feasibility issues).
- **Option 2 (Quasi-Infinite Horizon):** Use LQR cost for $l_f(x)$ and large N .
 - Don't enforce hard constraint $x_N \in \mathcal{X}_f$.
 - Relies on N being "long enough" to capture dynamics.
 - *Most common in walking robots and manipulators.*

6. Recursive Feasibility

Question: If I find a solution at time t , is a solution *guaranteed* to exist at $t + 1$?

Proof Idea:

- Suppose we have optimal trajectory $U_t^* = [u_0, u_1, \dots, u_{N-1}]$ at time t .
- Construct a **Candidate Solution** for time $t + 1$:

$$\tilde{U}_{t+1} = [u_1, u_2, \dots, u_{N-1}, \underbrace{\kappa_f(x_N)}_{\text{Terminal Control}}]$$

- If $\mathcal{X}_f$ is **Control Invariant**, this candidate is valid.
- Since a valid solution exists, the optimizer will find it (or a better one).

$\therefore$ **Control Invariance of $\mathcal{X}_f \implies$ Recursive Feasibility.**

7. Computational Speed: Warm Start

MPC requires solving an NLP in milliseconds (e.g., 1kHz loop).

Warm Start Strategy: Do not start the optimization from scratch (random guess or zero). Initialize the solver with the **shifted** previous solution:

$$U_{init}^{(t+1)} = [u_1^*, u_2^*, \dots, u_{N-1}^*, u_{N-1}^*]$$

Real-Time Iteration (RTI):

- Don't iterate to full convergence.
- Perform just **one SQP (Sequential Quadratic Programming) iteration** per time step.
- The feedback correction comes from the next time step's measurement.

8. Summary

- **MPC** turns optimal control into a feedback policy by repeatedly solving finite-horizon problems.
- **Stability** is not guaranteed by default with finite N . We need a proper **Terminal Cost** l_f and **Terminal Constraint** $\mathcal{X}_f$.
- **Recursive Feasibility** relies on the invariance of the terminal set.
- **Practical MPC** uses Warm Starting and sometimes accepts sub-optimal solutions (RTI) to meet real-time constraints (1ms).

Next: We will look at how to handle **Uncertainty** (Robust/Stochastic MPC).